

AI Search Framework: Version 3 Empirical Pilot Benchmark Report

Genuine All-SLM Local Pipeline (≤3.2B) vs. Monolithic Frontier Baseline (120B)

Date: September 14, 2026 | **Status:** Pilot Benchmark Completed, Audited & Diagnosed

Hardware: Local NVIDIA GeForce RTX 3050 6GB Laptop GPU (Vulkan Acceleration) + Groq LPU Cloud API

Integrity: Hard Rule 13 Pre-Flight Distinctness Verified | **Held-Out Checksum:**
c15452b4e421829d49cb8f0dbe4c8803ecb507402e5c6427200246fc681202b6 (100% Sealed)

1. Executive Summary & Plain-Language Scope

This report delivers the authentic, empirical results of the **Version 3 (v3) Pilot Benchmark** for the AI Search Framework. Following an audit that invalidated prior proxy runs, this benchmark strictly enforces **Hard Rule 13**, ensuring 100% distinct model identities across all components.

Scope & Host Constraints (Reported Plainly to Mentor):
Due to current cloud API landscape realities where third-party free ≥30B endpoints (OpenRouter, SambaNova, Google AI Studio Pro, Hugging Face Serverless) are paywalled with no free tier, **this benchmark empirical comparison is scoped to a single monolithic baseline: openai/gpt-oss-120b (120B) on Groq LPU.** To thoroughly test multi-domain specialization at zero dollar spend, the local pool was expanded on the RTX 3050 to **4 distinct open-weight SLMs (≤3.2B)** covering coding, mathematics, reasoning, retrieval, and synthesis.

Primary Empirical Finding:
The all-≤3.2B SLM pipeline loses decisively to the 120B monolithic baseline across all complexity tiers, with the margin widening as complexity increases: **9.4% aggregate win rate (3 Wins / 29 Losses / 0 Ties)** across 32 symmetric double-blind trials. Single-domain win rate (**12.5%**) was virtually identical to two-domain (**12.5%**), while compound DAG queries yielded **0.0%**. The architecture demonstrates competitive capability *only* in narrow, constrained algebraic derivation and formal deductive logic (`V3_SD_MATH_01` , `V3_SD_FORM_01`); it does **not** exhibit general single-domain competitiveness.

2. System Architecture & Model Pinning Roster

Component	Logical Role	Pinned Model Identifier	Host / Engine	Parameter Scale
Decomposer	Query Decomposition & DAG Generation	llama3.2:3b	Local Vulkan GPU	3.21B
Specialist 1	Coding, Structured Data, Systems Ops	qwen2.5-coder:3b	Local Vulkan GPU	3.09B
Specialist 2	Mathematics & Formal Reasoning	deepseek-r1:1.5b	Local Vulkan GPU	1.54B
Specialist 3	Retrieval QA & Documentation	qwen2.5:1.5b	Local Vulkan GPU	1.54B
Aggregator	Creative Synthesis & Aggregation	llama3.2:3b	Local Vulkan GPU	3.21B
Baseline	Monolithic Comparative Model	openai/gpt-oss-120b	Groq LPU Cloud API	120B
Judge	Double-Blind Pairwise Evaluator	qwen/qwen3.8-27b	Groq LPU Cloud API	27B

3. Empirical Results by Complexity Tier (32 Symmetric Trials)

Complexity Tier	Query Prefix	Trials	SLM Wins	120B Wins	Win Rate	Primary Differentiator Cited by Judge
Single-Domain (SD)	V3_SD_*	16	2	14	12.5%	Correctness (12) / Completeness (3) / Coherence (1)
Two-Domain (TD)	V3_TD_*	8	1	7	12.5%	Correctness (7) / Completeness (1)
Compound DAG (CD)	V3_CD_*	8	0	8	0.0%	Correctness (8 / 8 = 100%)
Aggregate Benchmark	All Tiers	32	3	29	9.4%	Correctness (27 / 32 = 84.4%)

Positional Agreement: 13 of 16 query pairs (**81.25%**) yielded identical outcomes across forward and swapped presentations, confirming high evaluation stability.

4. Empirical Reality: Response Length Disproves "Aggregator Over-Compression"

In v2, the working hypothesis was that SLMs lost due to aggregator over-compression (~249 chars vs 3,500 chars). The v3 empirical data **disproves this hypothesis entirely**:

Query ID	Complexity Tier	SLM Output Chars	120B Baseline Chars	Length Ratio (SLM/120B)	Judge Differentiator Cited
V3_CD_01	Compound DAG	8,500 chars	5,542 chars	1.53x (+53.4%)	correctness (both orders)
V3_CD_21	Compound DAG	6,055 chars	5,838 chars	1.04x (+3.7%)	correctness (both orders)
V3_CD_41	Compound DAG	5,882 chars	5,557 chars	1.06x (+5.9%)	correctness (both orders)
V3_CD_61	Compound DAG	4,752 chars	5,465 chars	0.87x (-13.0%)	correctness (both orders)
Mean (Compound)	—	6,297 chars	5,600 chars	1.12x (+12.4% longer)	Correctness (100% of trials)
V3_TD_11	Two-Domain	9,451 chars	7,846 chars	1.20x (+20.5%)	correctness (both orders)
V3_TD_31	Two-Domain	12,163 chars	6,584 chars	1.85x (+84.7%)	correctness (both orders)

Key Finding on Judge Behavior: In 8 out of 8 compound trials, the judge penalized **factual hallucinations and code invalidity (100% correctness)**. In multiple trials, the judge explicitly noted that the **120B baseline was truncated / cut off**, but still awarded it the win because the SLM pipeline's voluminous text contained fatal factual confabulations and broken syntax.

5. Intermediate Subtask Forensics: Verbatim Specialist Hallucinations

To determine where errors originated, raw subtask outputs were extracted from `results/v3_pilot/pipeline_logs/` prior to aggregation:

1. RFC Mislabeling & Generation Looping (`qwen2.5:1.5b` — Node 1 of `v3_CD_21`)

```
"2. RFC 2407: 'Security Architecture for the Internet Protocol (IP) - Extensions for TLS'
3. RFC 2409: 'Security Architecture for the Internet Protocol (IP) - Extensions for IKE'
...
7. RFC 2426: 'Security Architecture for the Internet Protocol (IP) - Extensions for ESP and AH - Extensions for ESP and AH'
8. RFC 2427: 'Security Architecture for the Internet Protocol (IP) - Extensions for ESP and AH - Extensions for ESP and AH'"
```

Diagnostic: Total parametric confabulation. RFC 2407 is ISAKMP Domain of Interpretation (not TLS); RFC 2426 is vCard MIME. The 1.5B model repeated hallucinated string suffixes in an ungrounded generation loop.

2. Kernel Socket Confabulation (`deepseek-r1:1.5b` — Node 2 of `v3_CD_21`)

```
"The original socket model used a specific structure, which has been replaced by a more modern approach. The new model uses a pointer to the socket instead of the socket ID, which can be manipulated, leading to cross-sockets or cross-kernel attacks... makes it easier for attackers to inject into systems running on different kernels."
```

Diagnostic: Pseudo-technical hallucination. Linux sockets are user-space file descriptors backed by kernel `struct socket`. "Cross-kernel attacks between different kernels via socket pointers" is physically nonsensical.

3. `virtualenv` as Security Sandbox (`qwen2.5-coder:3b` — Node 3 of `v3_CD_21`)

```
"Below is a basic example using a virtual environment to create a sandboxed Python runtime:
pip install virtualenv && virtualenv my_sandboxed_env && source my_sandboxed_env/bin/activate
Isolation: The virtual environment provides a basic level of isolation. However, it does not prevent code injection or other advanced attacks."
```

Diagnostic: Shallow tutorial matching. Conflates Python dependency management with security isolation (zero `seccomp`, zero `cgroups`, zero namespaces).

4. PDE Matrix Omission & Phantom `data.csv` (`qwen2.5-coder:3b` — Nodes 2 & 3 of `v3_CD_41`)

```
"Step 2: Prepare your data. Assume you have a CSV file named data.csv .
data = pd.read_csv('data.csv') → data.to_parquet('output.parquet') "
```

Diagnostic: The model failed to implement sparse Kronecker matrix math for the PDE, and completely decoupled from Node 1's simulation data, inventing a fictitious `data.csv` file on disk.

6. Routing Defect vs. Genuine Capability Ceiling Analysis

Observed Failure Mode	Assigned Specialist	Routing Validity	Root Cause Analysis	Fixable by Routing?
RFC Mislabeling	qwen2.5:1.5b	Appropriate (retrieval_qa)	Parametric knowledge sparsity in a 1.5B model without external RAG/search.	NO
Kernel Socket Confabulation	deepseek-r1:1.5b	Suboptimal (mapped to formal_reasoning)	Alternative in pool was qwen2.5-coder:3b , which also failed systems isolation.	NO
virtualenv as Sandbox	qwen2.5-coder:3b	Appropriate (coding)	Small coding models reproduce frequent tutorial templates rather than OS primitives.	NO
Missing Kronecker Matrix Math	qwen2.5-coder:3b	Appropriate (science_tech)	Insufficient scientific computing pre-training density for sparse Kronecker Laplacians.	NO
Phantom data.csv Disconnect	qwen2.5-coder:3b	Appropriate (structured_data)	Semantic dependency loss across DAG stages; falls back to generic pandas snippets.	NO

Diagnostic Verdict: This is **not a routing defect**. The TaskColorer directed tasks to designated specialists. The failures represent the **fundamental capability ceiling of ≤3.2B models** on precision-critical engineering tasks requiring exact parametric recall, deep systems knowledge, and multi-hop state retention.

7. Research Implications: Direct Answer to RQ3 & RQ4 at ≤5B Cap

- RQ3 (Quality Parity):** An all-≤3.2B decomposed SLM pipeline **cannot match a monolithic 120B model** on complex technical queries (9.4% overall, 0.0% on compound queries). Frontier LLMs maintain overwhelming advantages in cross-domain synthesis and domain correctness.
- RQ4 (Failure Modes):** The primary failure mode is **specialist parametric confabulation and dependency decoupling**, *not* aggregator over-compression. Errors in early DAG nodes cascade and amplify in downstream stages.
- Validation of Mentor's Intuition:** The mentor's skepticism regarding the ≤5B constraint is strongly supported by empirical logs. Small models lack the parametric capacity for complex technical synthesis without external tooling/RAG.

8. Efficiency Profile & Resource Costs

System	Architecture	Hardware / Host	Avg Latency	Avg Output Length	Total Spend
SLMPipeline_v3	4 Local SLMs (≤3.2B)	NVIDIA RTX 3050 6GB (Vulkan)	173.30s	6,590 chars	\$0.00
gpt_120b Baseline	Monolithic (120B)	Groq LPU Cloud API	4.65s	5,757 chars	\$0.00

9. Decision Point for Mentor Review (Pathway A vs. Pathway B)

It is critical to note that this pilot specifically tested models in the **1.5B–3.2B range** — the bottom tier of the mentor's authorized ≤5B constraint. It has **not yet tested anything in the 3.8B–5.0B range**, which remains strictly within the original authorized scope. The decision before the mentor is:

- Pathway A (Accept & Document ≤5B Failure):** Accept the 9.4% overall (0% on compound) pilot result as the definitive answer for lightweight decomposed SLMs, concluding that small models in the 1.5–3.2B range suffer from severe parametric sparsity on precision-critical engineering tasks.
- Pathway B (Test the Actual ≤5B Ceiling First — Within Authorized Range):** Before concluding that the *entire* authorized size class fails, run one small, cheap pilot at the true ≤5B boundary — evaluating true ≤5B models (such as Phi-3.5-mini at 3.8B combined with any viable 4–5B specialists) to determine whether the additional 1–2B parameters provide the threshold capacity needed for protocol recall and code synthesis, without changing the original ≤5B constraint.
- Separate Note on ≤8B (Constraint Modification):** Only if the mentor explicitly wishes to test whether 7B/8B models (e.g., qwen2.5-coder:7b , llama-3.1:8b) overcome this ceiling would we request formal approval to relax the constraint from ≤5B to ≤8B. We do not propose exceeding the authorized ≤5B cap unilaterally.
- Held-Out Partition Preserved:** All 160 queries in data/v3_queries_held_out.json remain sealed under SHA256 c15452b4...

Empirical Review Pack: September 14, 2026 | AI Search Framework | Repository: dixitabhi1/SLM_PROJECT